Automatic License Plate Recognition under Different Imaging Conditions

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Introduction:

Automatic License Plate Recognition is a computer vision technology used to identify vehicle registration plates from images. A typical recognition system first captures an image of a vehicle, identifies the license plate region, and then uses a Convolutional Neural Network (CNN) model to recognize the characters on the plate.

However, the performance of an AI recognition system does not only depend on the recognition algorithm, but also on the quality and conditions of the image supplied to it. A highly accurate recognition model may still fail when a plate is captured at low resolution, affected by motion blur, viewed from an unfavorable camera angle, or being exposed to improper illumination. The reliability of this process therefore depends strongly on the conditions under which the input image is captured. This raises an important practical question: **what kinds of image capturing conditions give an AI system the greatest probability of correctly recognizing a license plate? How to optimize it given that there is existing constraint?**

The main purpose of this paper is to construct a mathematical model analyzing different imaging conditions, which of them gives the highest conditional probability of successful recognition under five measurable factors: image resolution R, motion blur B, horizontal camera angle θ_h, vertical camera angle θ_v and illumination L. The mathematical framework is developed to understand which factors have the greatest influence on AI recognition and how they should be balanced when ideal conditions cannot all be achieved simultaneously.

To make the model experimentally grounded, images from the Chinese City Parking Dataset (CCPD2019) license plate dataset are used to train and evaluate a CNN recognition model. A proportionately stratified sample of 3,000 images is then used as the experimental dataset, and the recognition outcomes are used to estimate the coefficients of the mathematical probability function. In this way, the project connects mathematical modelling with a real AI recognition problem rather than relying entirely on assumed parameter values.

A major aim of the project is to apply the multivariable calculus and optimization methods studied in class to a real-world problem. Partial derivatives are used to measure the sensitivity of recognition probability to changes in each of the five factors. Second-order

partial derivatives and the Hessian matrix are then used to examine the interactions within five factors, providing a deeper understanding of how the recognition probability changes around different operating conditions. Finally, Lagrange multipliers are used to formulate a constrained optimization problem in which improvements to image capturing conditions must be balanced with a limited budget.

Section 1: Set up the model for four influencing factors

Our dependent variable P is the conditional probability of successful recognition given certain imaging conditions. Therefore, $0 \leq P \leq 1$, where $P = 1$: perfect recognition; $P = 0.95$: 95% accuracy; $P = 0.3$: poor recognition; $P = 0$: impossible to recognize.

Four factors are: Resolution (unit: pixels); Motion blur (unit: pixels); Illumination (unit: lux); Camera angle (unit: degree).

(a) Resolution

The resolution variable normally represented the resolution of the entire captured image. However, the overall image resolution does not directly determine whether a license plate can be recognized. For example, a high-resolution image may still contain a very small license plate if the vehicle is far away from the camera. Therefore, I define the resolution variable as the **effective license plate resolution**, namely the number of pixels occupied by the detected license plate.

Assume the detected license plate has width w_p pixels and height h_p pixels. Then we define

$$R = \frac{w_p h_p}{WH},$$

where R denotes the effective resolution of the licence plate rather than the entire image

Consider

$$P_R(R) = \frac{1}{1 + e^{-a(R-R_0)}}.$$

Recognition improves rapidly at first. Further increases contribute progressively less to the overall recognition performance. It naturally captures the phenomenon of diminishing returns.

where

- R_0 is the critical resolution at which the contribution of resolution reaches 50%
- a>0

- $\lim_{R\to\infty} P_R(R) = 1$,
- W = image width,
- H = image height.

(b) Motion Blur

An appropriate function to model motion blur is exponential decay. The effect of motion blur is also similar to diminishing returns. It affects the recognition a lot at first, further motion blur only provides marginal effects.

$$P_B(B) = e^{-kB}.$$

(c) Camera Angle — Horizontal and Vertical Tilt

Define $\theta_h =$ horizontal tilt angle, and $\theta_v =$ vertical tilt angle.
Perspective distortion of a license plate can occur in two independent directions.

The recognition accuracy reaches its maximum when the camera faces the license plate directly, namely

$$\theta_h = \theta_v = 0.$$

Since clockwise and anticlockwise rotations of equal magnitude should have identical effects, the contribution of each angle is modelled by its squared magnitude.

As the angle increases, accuracy decreases.

The decrease should be smooth. Small rotations (2° → 5°) should not suddenly cause failure.

Eventually accuracy becomes extremely small.

Consider

$$P_\theta(\theta_h, \theta_v) = e^{-k_h\theta_h^2 - k_v\theta_v^2}.$$

Where $k_h > 0$ measures the sensitivity to horizontal camera tilt, while $k_v > 0$ measures the sensitivity to vertical camera tilt.

This formulation also allows the two directions to have different sensitivities. $k_h > k_v$ indicates that horizontal misalignment has a greater impact on recognition performance than vertical misalignment.

(d) Illumination

If it is dark, recognition is poor. If it is moderate, everything is clear, recognition is excellent. If it is very bright, like sun glare, it is likely for overexposure to happen, recognition decreases again.

Consider

$$P = e^{-\frac{(L-L_0)^2}{2\sigma^2}}$$

where L_0 is the ideal brightness, suppose it is 50.

Brightness	Probability
0	0.04
20	0.33
40	0.88
50	1.00
60	0.88
80	0.33
100	0.04

- Symmetry around the optimum
- Fast decay, which means performance falls gradually near the optimum but more rapidly as conditions become extreme

This perfectly captures the idea that both insufficient and excessive illumination reduce recognition performance.

After the construction of four factors, suppose each factor contributes **independently.**

Then

$$P = P_R P_B P_L P_\theta.$$

Substitute every function.

$$P(R, B, L, \theta_h, \theta_v) = \frac{1}{1 + e^{-a(R-R_0)}} e^{-k_1 B} e^{-\frac{(L-L_0)^2}{2\sigma^2}} e^{-k_h \theta_h^2 - k_v \theta_v^2}.$$

Our dependent variable P is finally written as

$$P = P(Y = 1 \mid R, B, L, \theta_h, \theta_v),$$

Which is a conditional probability, where $Y = 1$ denotes that the licence plate is recognised correctly.

Then the coefficients of the combined function are worked out from the experimental data as stated in Step 2.

Section 2: Data Collection and Stratified Sampling

The experimental data used in this paper are obtained from the **Chinese City Parking Dataset (CCPD2019)** from GitHub. 《*Towards End-to-End License Plate Detection and Recognition: A Large Dataset and Baseline*》 In particular, CCPD_Base subset is used for parameter estimation. It contains 95,774 license plate images under a range of real-world imaging conditions. An important feature of CCPD is that information describing each image is encoded directly in its filename. This includes the relative area occupied by the license plate, horizontal and vertical camera tilt, brightness, blurriness, and the ground-truth license plate number. These annotations correspond closely to the explanatory variables used in this paper's mathematical model. Hence, each image provides the imaging conditions together with the license plate number. However, using all 95,774 images would be computationally unnecessary for the experimental stage. Since these extreme conditions are particularly important for estimating the sensitivity parameters of the model, a **multivariate stratified sampling procedure** is used to select a representative experimental sample of 3,000 images.

For each of the five explanatory variables $S_i = (R_i, B_i, L_i, \theta_{h,i}, \theta_{v,i})$, the empirical distribution is divided into three approximately equal-frequency categories: Low, Medium, High, using the 33rd and 67th percentiles of the full CCPD_Base dataset as the approximate dividing points. Each image can therefore be classified according to five categories simultaneously. All $3^5 = 243$ combinations are represented in the extracted CCPD_Base data. The 3,000 experimental images are then allocated across these strata using **proportionate stratified sampling**. Thus, strata that occur frequently in the original dataset contribute proportionally more observations to the experimental sample, while each occupied stratum is represented by at least one observation. Within each stratum, images are selected randomly.

The resulting 3,000-image sample closely preserves the overall characteristics of the complete dataset. For example, the mean blurriness is 136.97 in the complete dataset and 134.14 in the sample, while the mean horizontal tilt is 4.99 and 4.81, respectively. This shows that the reduced sample preserves as many original observations as possible, rather than concentrating only on the most common imaging conditions.

Section 3: Build up the Recognition Model using CNN

To generate the binary recognition outcome required for parameter estimation, a separate recognition model is constructed using the CCPD2019 data. The purpose of this model is not to replace the mathematical probability model, but to provide an experimental mechanism for determining whether each sampled license plate is recognized successfully.

Before recognition, the license plate region of each image is extracted using the simple bounding-box annotation supplied by CCPD. Only simple cropping is performed; no geometric correction is applied. This is important because the effects of resolution, blur, illumination, and camera angle needed to remain present in the images used for the experiment.

Each cropped license plate was converted to grayscale and resized to a common input size of

32 * 160 pixels. Pixel intensities were then normalized before being passed to the neural network.

The stratified sample of 3,000 images I developed in the previous section is reserved as the final experimental dataset.

The remaining images are separated into training and validation sets. Specifically, 30,000 images were used for training and 5,000 images for validation.

The recognition model is formulated as a fixed seven-position CNN classifier because every target plate contains 7 characters only. Instead of predicting a variable-length sequence, the network directly predicts $(\hat{c}_1, \hat{c}_2, \hat{c}_3, \hat{c}_4, \hat{c}_5, \hat{c}_6, \hat{c}_7)$. The first output classifies the province character, the second output classifies the alphabetic character, and the remaining five outputs classify alphanumeric characters.

During the training process, I use a separate cross-entropy loss for each character position. The total training loss is represented by $L = \sum_{j=1}^{7} L_j$, where Lj is the cross-entropy loss associated with character position j.

A preliminary model is first trained on 10,000 images to verify that the recognition pipeline worked correctly. After five epochs, the model achieves approximately 92.99% character level accuracy and 62.25% exact plate accuracy. The learned model is then further trained using 30,000 training images and 5,000 validation images with a reduced learning rate. The final validation results are character accuracy= 97.90% and exact plate accuracy= 88.82%.

The trained model was then frozen and applied to the 3,000 previously unseen stratified experimental images.

Of the 3,000 experimental images, 2,660 were recognized exactly correctly and 340 were

incorrect, giving an overall experimental accuracy of 88.67%.

Section 4: Parameter Estimation Using Bernoulli Maximum Likelihood

Each observation can therefore be represented as a binary outcome according to whether the complete license plate was successfully recognized.

Let the ground-truth license plate associated with image be: s_i, and let the license plate predicted by algorithm be: $\hat{s}_i$. The recognition outcome is then represented by a binary response variable $Y_i = 1, \quad \hat{s}_i = s_i$ meaning that the license plate is recognized correctly, or $Y_i = 0, \quad \hat{s}_i \neq s_i$ meaning that the license plate is recognized incorrectly. Since there are only two possible outcomes, the recognition result can be modelled using **a Bernoulli Distribution**: $Y_i \sim \text{Bernoulli}(P_i)$, where P_i is the conditional probability that the ith licence plate is recognized correctly under a specific imaging condition. Therefore, each final experimental observation can be represented as: $(R_i, B_i, L_i, \theta_{(h,i)}, \theta_{(v,i)}, Y_i)$ and the parameter vector can be written as: $\beta = (a, R_0, k_1, L_0, \sigma, k_h, k_v)$.

For a Bernoulli random variable, the probability mass function can be written as $P(Y_i = y_i) = P_i^{y_i}(1 - P_i)^{1-y_i}$.

These parameters will subsequently be estimated using **Maximum Likelihood Estimation**, where the observed values of are used to determine which combination of parameter values makes the experimental recognition outcomes most probable.

Assuming that the 3,000 observations are treated as independent conditional on their measured factors, the likelihood of observing the complete experimental dataset is therefore $\mathcal{L}(\boldsymbol{\beta}) = \prod_{i=1}^{3000} P_i^{Y_i}(1 - P_i)^{1-Y_i}$.

Hence, $\beta = \underset{\boldsymbol{\beta}}{\text{argmax}}\ \mathcal{L}(\boldsymbol{\beta})$.

Directly multiplying 3,000 probabilities can lead to the number to be extremely small. Therefore, I take the logarithm of the likelihood function. Let $\ell(\boldsymbol{\beta}) = \ln \mathcal{L}(\boldsymbol{\beta})$, since the logarithm function is strictly increasing, maximizing the log-likelihood gives the same parameter estimates as maximizing the original likelihood.

Applying the logarithm to the product gives: $\ell(\boldsymbol{\beta}) = \sum_{i=1}^{3000} [\, Y_i \ln(P_i) + (1 - Y_i) \ln(1 - P_i)]$

The parameters are then determined by numerically maximizing this log-likelihood subject to the physically meaningful conditions: $a > 0,\ k_1 \geq 0,\ \sigma > 0,\ k_h \geq 0,\ k_v \geq 0$.

The resulting Maximum Likelihood estimates are:

(Note: This is run by computer calculations.)

$$\hat{a} = 0.01496,$$

$$\hat{R}_0 = 2.00,$$

$$\hat{k}_1 = 1.73 \times 10^{-4},$$

$$\hat{L}_0 = 28.00,$$

$$\hat{\sigma} = 399.62,$$

$$\hat{k}_h = 4.62 \times 10^{-4},$$

$$\hat{k}_v = 1.86 \times 10^{-4}.$$

Therefore, the recognition probability model becomes:

$$P(R, B, L, \theta_h, \theta_v) = \frac{1}{1+e^{-0.01496(R-2)}} \; e^{-0.000173B} \; e^{-\frac{(L-28)^2}{2(399.62)^2}} \; e^{-0.000462\theta_h^2 - 0.000186\theta_v^2}.$$

There are some findings from the resulting Maximum Likelihood estimates:

- $\hat{\sigma} = 399.62$ indicates that the graph of the illumination function is almost flat, within the brightness range represented in this CCPD sample, brightness has relatively little influence on the recognizer. When $L = 117$, the probability contributed by the illumination function is already 0.976.
- When $R = 393$, the probability contributed by the illumination function is already 0.997. This shows that most license plates in our selected sample already have sufficient effective resolution for the CNN.
- Blur has a somewhat larger effect for the overall probability function. If blur increases by 1000 units, the model predicts roughly a 15.9% reduction in the blur component.
- As $\hat{k}_h = 4.62 \times 10^{-4}$; $\hat{k}_v = 1.86 \times 10^{-4}$, $\hat{k}_h$ is approximately 2.5 times compared with $\hat{k}_v$. Therefore, the recognition system is approximately 2.5 times more sensitive to horizontal tilt than to vertical tilt.

Section 5: Sensitivity Analysis using Partial Derivatives and Hessian Matrix

(a) Resolution

$$P_R = \frac{1}{1 + e^{-a(R-R_0)}}$$

$$\boxed{\frac{\partial P}{\partial R} = \frac{dP_R}{dR} P_B P_L P_\theta.}$$

Let

$$u = a(R - R_0)$$

$$P_R = \frac{1}{1 + e^{-u}}.$$

Using the chain rule,

$$\frac{dP_R}{dR} = \frac{dP_R}{du} \cdot \frac{du}{dR}.$$

$$\frac{du}{dR} = a.$$

$$\frac{dP_R}{du} = P_R(1 - P_R)$$

$$\boxed{\frac{\partial P}{\partial R} = aP_R(1 - P_R)P_B P_L P_\theta.}$$

Since $P = P_R P_B P_L P_\theta$, we can simplify to

$$\boxed{\frac{\partial P}{\partial R} = aP(1 - P_R).}$$

$$\frac{\partial P}{\partial R} = 0.01496P(1 - P_R)$$

Second derivative using the product rule gives:

$$\frac{\partial^2 P}{\partial R^2} = a\left[\frac{\partial P}{\partial R}(1 - P_R) - P\frac{\partial P_R}{\partial R}\right].$$

Since

$$\frac{\partial P}{\partial R} = aP(1 - P_R)$$

and

$$\frac{\partial P_R}{\partial R} = aP_R(1 - P_R),$$

Substitute both expressions:

$$\boxed{\frac{\partial^2 P}{\partial R^2} = a^2 P(1 - P_R)(1 - 2P_R).}$$

$$\frac{\partial^2 P}{\partial R^2} = (0.01496)^2 P(1 - P_R)(1 - 2P_R)$$

(b) Motion Blur

$$P_B = e^{-k_1 B}$$

$$\frac{\partial P}{\partial B} = P_R P_L P_\theta \frac{dP_B}{dB}.$$

Differentiate:

$$\frac{dP_B}{dB} = -k_1 e^{-k_1 B} = -k_1 P_B.$$

Hence

$$\boxed{\frac{\partial P}{\partial B} = -k_1 P.}$$

$$\frac{\partial P}{\partial B} = -0.000173P$$

Differentiate again:

$$\frac{\partial^2 P}{\partial B^2} = -k_1(-k_1 P).$$

Therefore

$$\boxed{\frac{\partial^2 P}{\partial B^2} = k_1^2 P.}$$

$$\frac{\partial^2 P}{\partial B^2} = (0.000173)^2 P$$

(c) Camera Angle

For the horizontal angle component:

$$P_{\theta_h} = e^{-k_h \theta_h^2}$$

The first derivative gives:

$$\frac{dP_{\theta_h}}{d\theta_h} = -2k_h\theta_h P_{\theta_h}$$

Thus, for the full function,

$$\frac{\partial P}{\partial \theta_h} = -2k_h\theta_h P$$

Substituting k_h = 0.000462 gives:

$$\frac{\partial P}{\partial \theta_h} = -0.000924\theta_h P$$

Second derivative using the product rule gives:

$$\frac{\partial^2 P}{\partial \theta_h^2} = -2k_h\left(P + \theta_h\frac{\partial P}{\partial \theta_h}\right)$$

$$\frac{\partial^2 P}{\partial \theta_h^2} = P\left(4k_h^2\theta_h^2 - 2k_h\right)$$

Substituting k_h = 0.000462 gives:

$$\frac{\partial^2 P}{\partial \theta_h^2} = P\left[4(0.000462)^2\theta_h^2 - 2(0.000462)\right]$$

For the vertical angle component:

$$P_{\theta_v} = e^{-k_v\theta_v^2}$$

Taking the first derivative, for the same reason:

$$\frac{\partial P}{\partial \theta_v} = -2k_v\theta_v P$$

Substituting $k_v = 0.000186$:

$$\frac{\partial P}{\partial \theta_v} = -0.000372\theta_v P$$

Second derivative using the product rule gives:

$$\frac{\partial^2 P}{\partial \theta_v^2} = -2k_v\left(P + \theta_v\frac{\partial P}{\partial \theta_v}\right)$$

Simplify:

$$\frac{\partial^2 P}{\partial \theta_v^2} = P(4k_v^2\theta_v^2 - 2k_v)$$

Substituting $k_v = 0.000186$:

$$\frac{\partial^2 P}{\partial \theta_v^2} = P[4(0.000186)^2\theta_v^2 - 2(0.000186)]$$

(d) Illumination

$$P_L = e^{-\frac{(L-L_0)^2}{2\sigma^2}}.$$

Let

$$v = -\frac{(L-L_0)^2}{2\sigma^2}.$$

Using the chain rule,

$$\frac{dP_L}{dL} = \frac{dP_L}{dv} \cdot \frac{dv}{dL}.$$

$$\frac{dP_L}{dv} = e^v = P_L,$$

$$\frac{dv}{dL} = -\frac{L-L_0}{\sigma^2}.$$

Therefore,

$$\boxed{\frac{dP_L}{dL} = -\frac{L-L_0}{\sigma^2}P_L.}$$

Substituting back into the full model,

$$\boxed{\frac{\partial P}{\partial L} = -\frac{L-L_0}{\sigma^2}P.}$$

Substituting $L_0 = 28$ and $\sigma = 399.62$:

$$\frac{\partial P}{\partial L} = -\frac{L-28}{(399.62)^2}P$$

Apply the product rule:

$$\frac{\partial^2 P}{\partial L^2} = -\frac{1}{\sigma^2}P - \frac{L-L_0}{\sigma^2}\frac{\partial P}{\partial L}.$$

Substitute

$$\frac{\partial P}{\partial L} = -\frac{L - L_0}{\sigma^2} P.$$

$$\boxed{\frac{\partial^2 P}{\partial L^2} = P\left(\frac{(L - L_0)^2}{\sigma^4} - \frac{1}{\sigma^2}\right).}$$

Substituting $L_0 = 28$ and $\sigma = 399.62$:

$$\frac{\partial^2 P}{\partial L^2} = P\left[\frac{(L - 28)^2}{(399.62)^4} - \frac{1}{(399.62)^2}\right]$$

After obtaining the first and second partial derivatives for each variable, the Hessian matrix is used to summarize the second-order behavior of the recognition probability function

The Hessian matrix is:

$$H(P) = \begin{pmatrix} \frac{\partial^2 P}{\partial R^2} & \frac{\partial^2 P}{\partial R \partial B} & \frac{\partial^2 P}{\partial R \partial L} & \frac{\partial^2 P}{\partial R \partial \theta_h} & \frac{\partial^2 P}{\partial R \partial \theta_v} \\ \frac{\partial^2 P}{\partial B \partial R} & \frac{\partial^2 P}{\partial B^2} & \frac{\partial^2 P}{\partial B \partial L} & \frac{\partial^2 P}{\partial B \partial \theta_h} & \frac{\partial^2 P}{\partial B \partial \theta_v} \\ \frac{\partial^2 P}{\partial L \partial R} & \frac{\partial^2 P}{\partial L \partial B} & \frac{\partial^2 P}{\partial L^2} & \frac{\partial^2 P}{\partial L \partial \theta_h} & \frac{\partial^2 P}{\partial L \partial \theta_v} \\ \frac{\partial^2 P}{\partial \theta_h \partial R} & \frac{\partial^2 P}{\partial \theta_h \partial B} & \frac{\partial^2 P}{\partial \theta_h \partial L} & \frac{\partial^2 P}{\partial \theta_h^2} & \frac{\partial^2 P}{\partial \theta_h \partial \theta_v} \\ \frac{\partial^2 P}{\partial \theta_v \partial R} & \frac{\partial^2 P}{\partial \theta_v \partial B} & \frac{\partial^2 P}{\partial \theta_v \partial L} & \frac{\partial^2 P}{\partial \theta_v \partial \theta_h} & \frac{\partial^2 P}{\partial \theta_v^2} \end{pmatrix}$$

Since the probability function is smooth, the mixed second derivatives satisfy:

$$\frac{\partial^2 P}{\partial x_i \partial x_j} = \frac{\partial^2 P}{\partial x_j \partial x_i}$$

The diagonal entries that have already been obtained are:

$$\frac{\partial^2 P}{\partial R^2} = a^2 P(1 - P_R)(1 - 2P_R)$$

$$\frac{\partial^2 P}{\partial B^2} = k_1^2 P$$

$$\frac{\partial^2 P}{\partial L^2} = P\left[\frac{(L - L_0)^2}{\sigma^4} - \frac{1}{\sigma^2}\right]$$

$$\frac{\partial^2 P}{\partial \theta_h^2} = P\left(4k_h^2\theta_h^2 - 2k_h\right)$$

$$\frac{\partial^2 P}{\partial \theta_v^2} = P(4k_v^2\theta_v^2 - 2k_v)$$

Since the Hessian matrix is symmetric, we only need 10 unique off-diagonal entries.

$$\frac{\partial^2 P}{\partial R \partial B} = -a k_1 P(1 - P_R)$$

$$\frac{\partial^2 P}{\partial R \partial L} = -\frac{a(L - L_0)}{\sigma^2} P(1 - P_R)$$

$$\frac{\partial^2 P}{\partial R \partial \theta_h} = -2a k_h \theta_h P(1 - P_R)$$

$$\frac{\partial^2 P}{\partial R \partial \theta_v} = -2a k_v \theta_v P(1 - P_R)$$

$$\frac{\partial^2 P}{\partial B \partial L} = \frac{k_1 (L - L_0)}{\sigma^2} P$$

$$\frac{\partial^2 P}{\partial B \partial \theta_h} = 2 k_1 k_h \theta_h P$$

$$\frac{\partial^2 P}{\partial B \partial \theta_v} = 2 k_1 k_v \theta_v P$$

$$\frac{\partial^2 P}{\partial L \partial \theta_h} = \frac{2 k_h \theta_h (L - L_0)}{\sigma^2} P$$

$$\frac{\partial^2 P}{\partial L \partial \theta_v} = \frac{2 k_v \theta_v (L - L_0)}{\sigma^2} P$$

$$\frac{\partial^2 P}{\partial \theta_h \partial \theta_v} = 4 k_h k_v \theta_h \theta_v P$$

The remaining ten off-diagonal entries are obtained from symmetry.

The Hessian matrix summarizes the curvature of the recognition probability with respect to all design variables simultaneously.

Beyond identifying the optimal operating point, the Hessian matrix provides insight into the local behavior of the recognition model. The diagonal second-order partial derivatives quantify the curvature of the probability function with respect to each design variable, revealing where diminishing returns occur and identifying which variables require precise control for engineers.

The mixed second-order partial derivatives describe interactions between variables, such as whether increasing image resolution changes the system's sensitivity to motion blur.

To examine the local behavior around the best illumination and camera alignment conditions, the conditions are: $L = 28$ and $\theta_h = \theta_v = 0$

$\frac{\partial^2 P}{\partial L^2} = -\frac{P}{(399.62)^2} < 0$, indicating that $L = 28$ represents a local maximum of the illumination component.

$\frac{\partial^2 P}{\partial \theta_h^2} = -2k_h P = -0.000924P < 0$; $\frac{\partial^2 P}{\partial \theta_v^2} = -2k_v P = -0.000372P < 0$, confirming local maxima when the license plate is viewed without horizontal or vertical angular deviation.

Furthermore, from the first derivatives: $\frac{\partial P}{\partial R} = aP(1 - P_R) > 0$ and $\frac{\partial P}{\partial B} = -k_1 P < 0$,

The probability of recognition increases monotonically with resolution and decreases monotonically with blur.

Therefore, there is no interior stationary point for R and B, they obtain their best values at the boundaries of the feasible region.

So, the gradient of the function is never zero. Consequently, the Hessian cannot be described as negative definite. The function does not exist an unconstrained local maximum.

Hence, we need some practical constraints, motivating the use of Lagrange Multipliers for constrained optimization in the last section.

Section 6: Constrained Optimization Using Lagrange Multipliers

Our objective is

$$\boxed{\max P\,(R, B, L, \theta)}$$

which means finding the camera settings that maximize the recognition probability.

A practical constraint is:

Given limited resources, how should resources be allocated among resolution, blur reduction, illumination control, and camera-angle improvement to maximize recognition probability?

Suppose we have an existing system; the budget could be constructed from the amount by which each factor is improved.

$c_R(R - R_0) + c_B(B_0 - B) + c_L(L_0 - L) + c_h(\theta_{h0} - \theta_h) + c_v(\theta_{v0} - \theta_v) = C$

Here C is the total available budget. The difference means for example, $B_0 - B$ is the amount of blur reduction improved.

Let the constraint be:

$g = C - c_R(R - R_0) - c_B(B_0 - B) - c_L(L_0 - L) - c_h(\theta_{h0} - \theta_h) - c_v(\theta_{v0} - \theta_v)$

Setting up $L = P(R, B, L, \theta_h, \theta_v) + \lambda g$

Differentiate with respect to all five variables respectively:

$\frac{\partial \mathcal{L}}{\partial R} = 0$, $\frac{\partial \mathcal{L}}{\partial B} = 0$, $\frac{\partial \mathcal{L}}{\partial L} = 0$, $\frac{\partial \mathcal{L}}{\partial \theta_h} = 0$, $\frac{\partial \mathcal{L}}{\partial \theta_v} = 0$; and we also have: $\frac{\partial \mathcal{L}}{\partial \lambda} = 0$.

Applying Lagrange multiplier rule, we obtain:

$$0.01496P(1 - P_R) = \lambda c_R$$

$$0.000173P = \lambda c_B$$

$$\frac{L-28}{(399.62)^2} P = \lambda c_L$$

$$0.000924\theta_h P = \lambda c_h$$

$$0.000372\theta_v P = \lambda c_v$$

One assumption is made in the calculation process: the displayed illumination equation assumes the starting illumination lies above the fitted optimum $L = 28$; if it lies below 28, the direction/sign of the illumination improvement must be reversed.

At the optimum, every pound spent should produce the same increase in recognition probability, regardless of whether it is spent on resolution, blur reduction, camera angle adjusting or lighting.

Since actual engineering cost data are not available in this project, a unique numerical optimal configuration is not calculated. Nevertheless, the Lagrange multiplier analysis establishes the general mathematical condition that any resource-efficient optimum must satisfy, completing the optimization stage of the model.